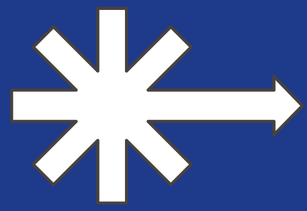
FINANCIAL DATA ANALYTICS

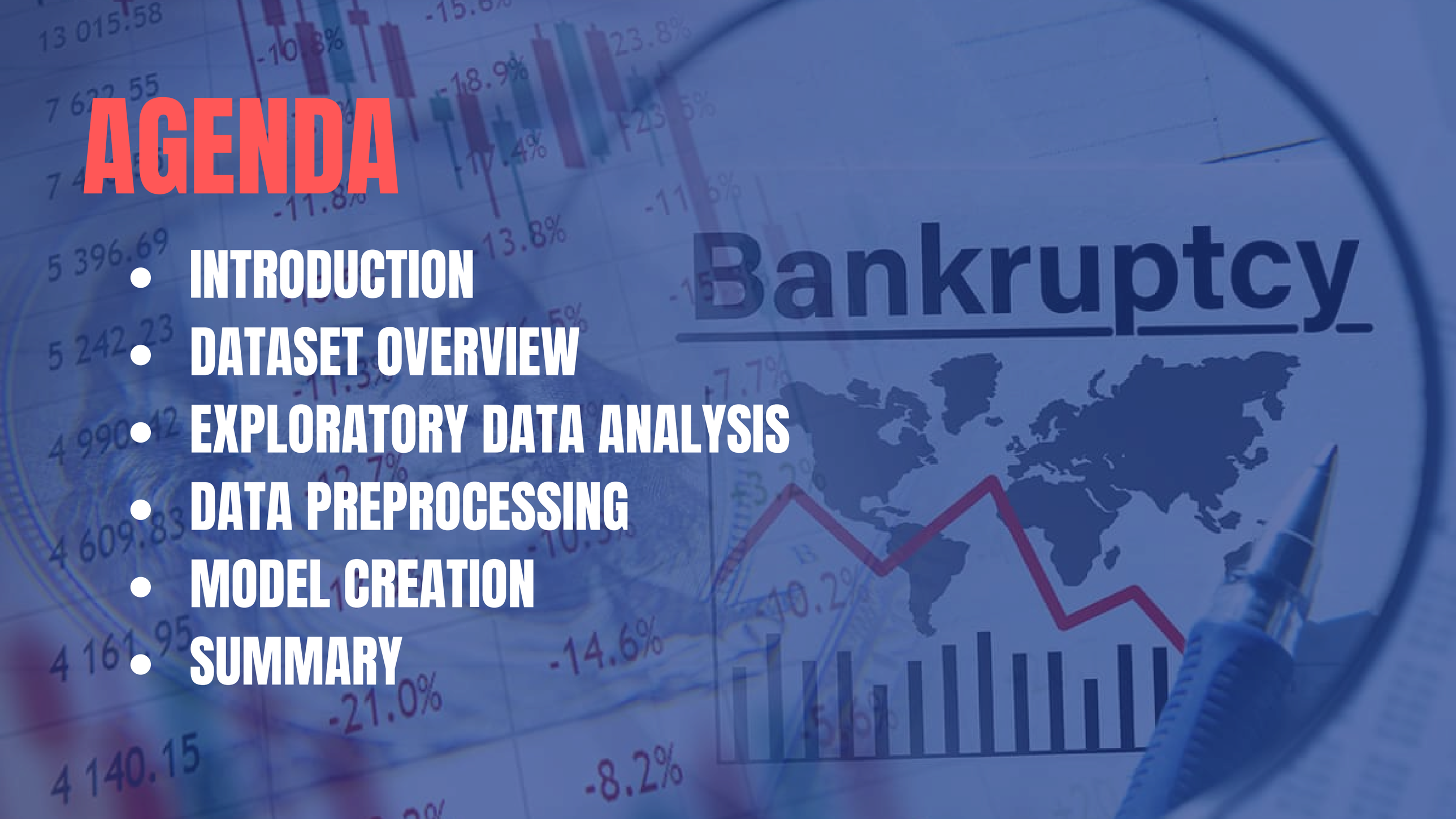
BANKRUPTCY PREDICTION



The Taiwanese Companies Dataset

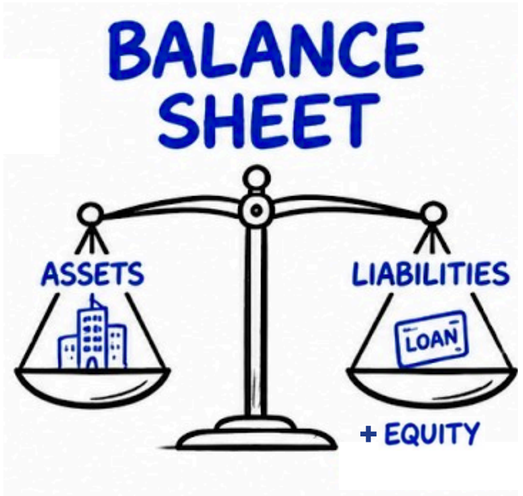
AGENDA

- INTRODUCTION
- DATASET OVERVIEW
- EXPLORATORY DATA ANALYSIS
- DATA PREPROCESSING
- MODEL CREATION
- SUMMARY



IS THIS COMPANY HEALTHY ?







WHY PREDICT BANKRUPTCY ?







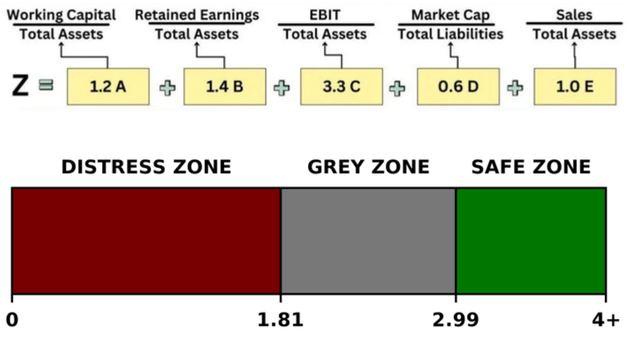


IMPACT ▷ creditors, investors, employees, suppliers, financial system

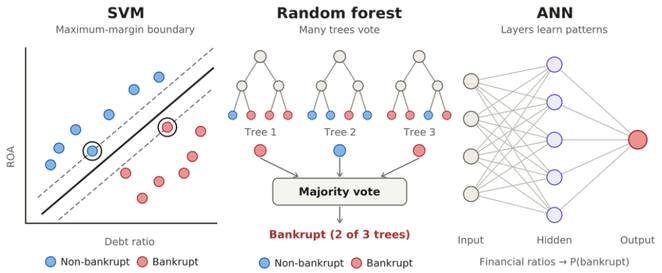
VALUE ▷ better lending decisions, portfolio risk management, early warning

FROM CLASSICAL STATISTICS TO ML

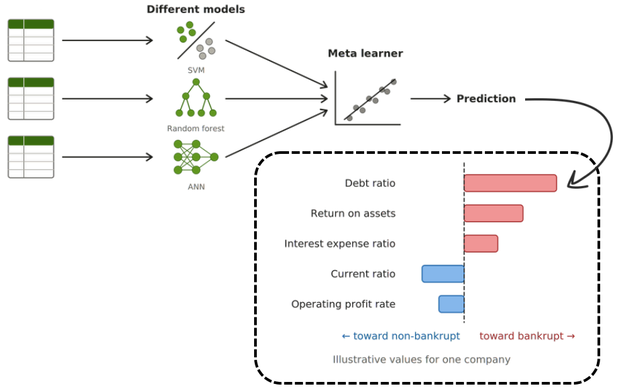
ALTMAN Z-SCORE



MACHINE LEARNING



ENSEMBLES +XAI



DATASET OVERVIEW

Bankruptcy data from the Taiwan Economic Journal for the years 1999–2009



Taiwanese Bankruptcy











Data Dimensions and Structure

Item	Value
Number of rows (companies)	6,819
Number of columns	96 (95 features + 1 target)
Float columns	93
Integer columns	3 (Bankrupt?, Liability-Assets Flag, Net Income Flag)
Target variable	**Bankrupt? (0 = not bankrupt, 1 = bankrupt)**

Features Examples

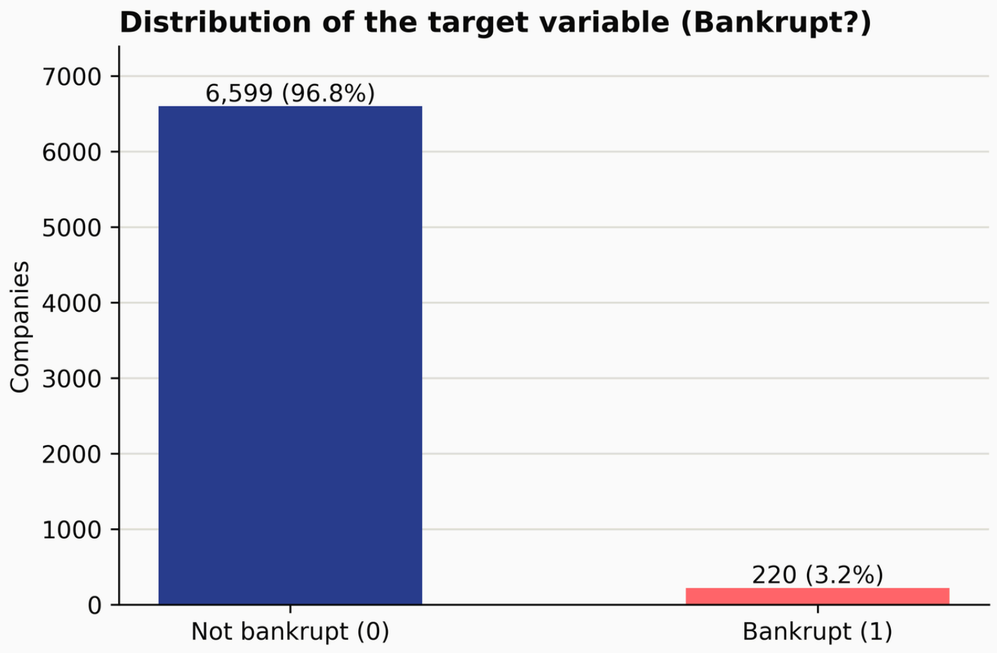
2. ROA(C) before interest and depreciation before interest
3. ROA(A) before interest and % after tax
4. ROA(B) before interest and depreciation after tax
5. Operating Gross Margin
6. Realized Sales Gross Margin

.
.
.

93. Degree of Financial Leverage (DFL)
94. Interest Coverage Ratio (Interest expense to EBIT)
95. Net Income Flag
96. Equity to Liability

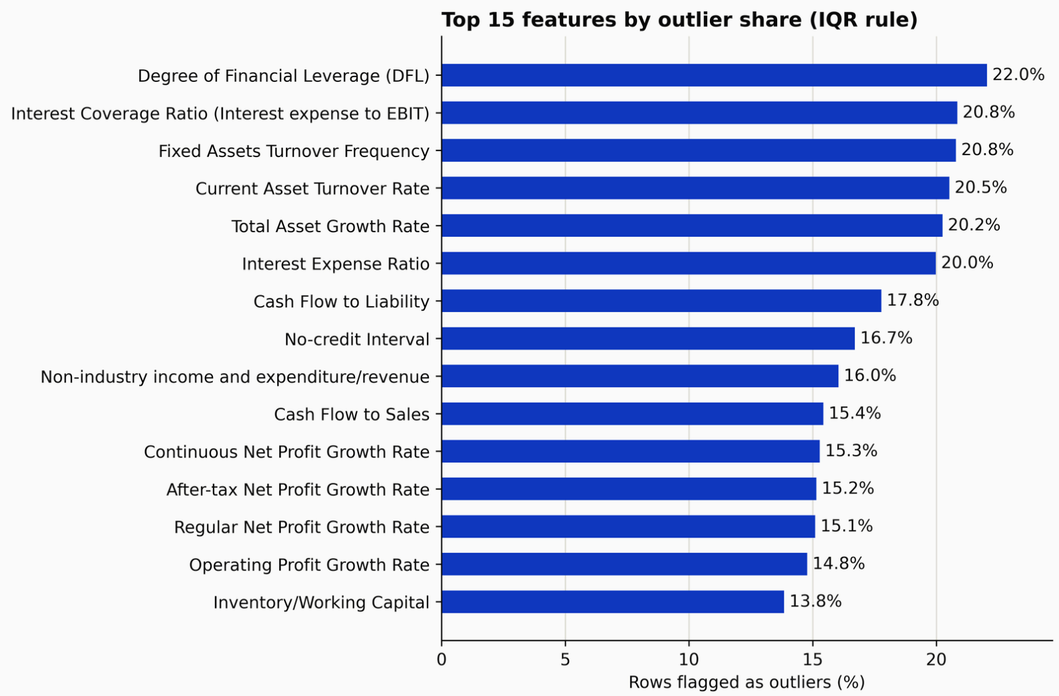
Data Quality Check

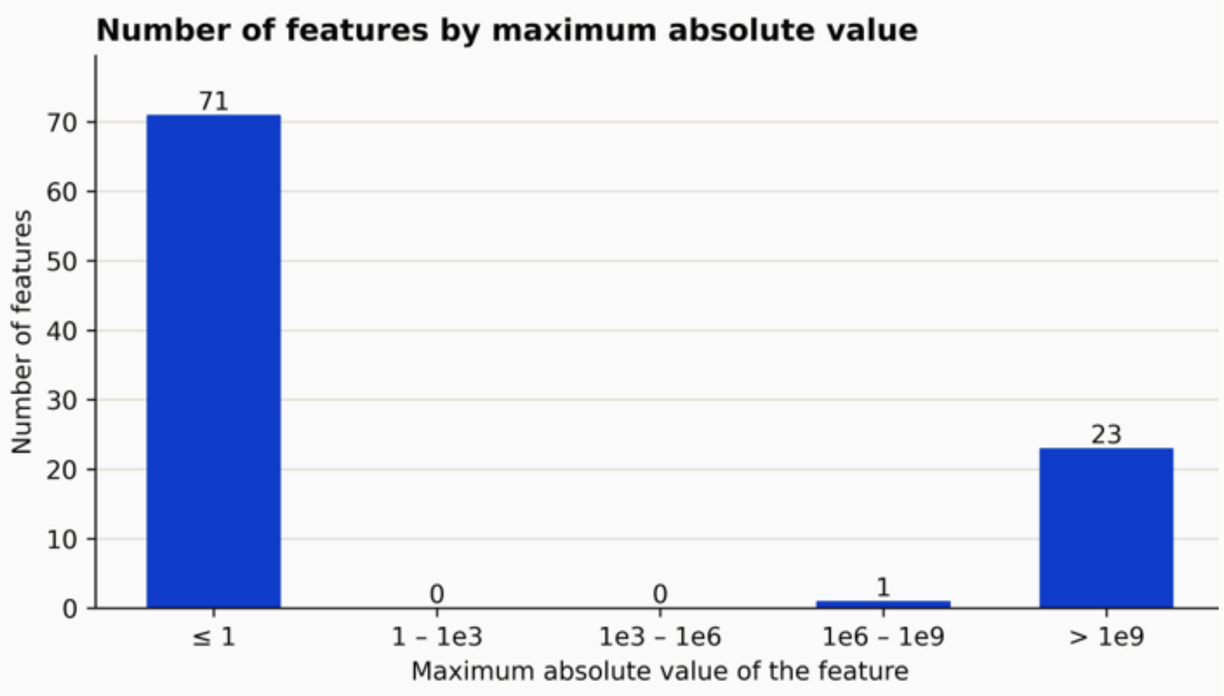
Check	Result
Missing values	0 in all columns
Duplicate rows	0
Infinite values	0
Constant columns	1 (Net Income Flag, which has the same value for every company)
Rows with at least one IQR outlier	6,361 of 6,819 (93.3%), including 219 of the 220 bankrupt companies



Clean data, no missing values or duplicates, but rare events: only 3.2% of companies went bankrupt.

OUTLIERS AND FEATURE SCALE





Outliers in up to 22% of rows 23 and features jump from 0-1 to over 1 billion, so we need scaling and outlier handling before modeling.

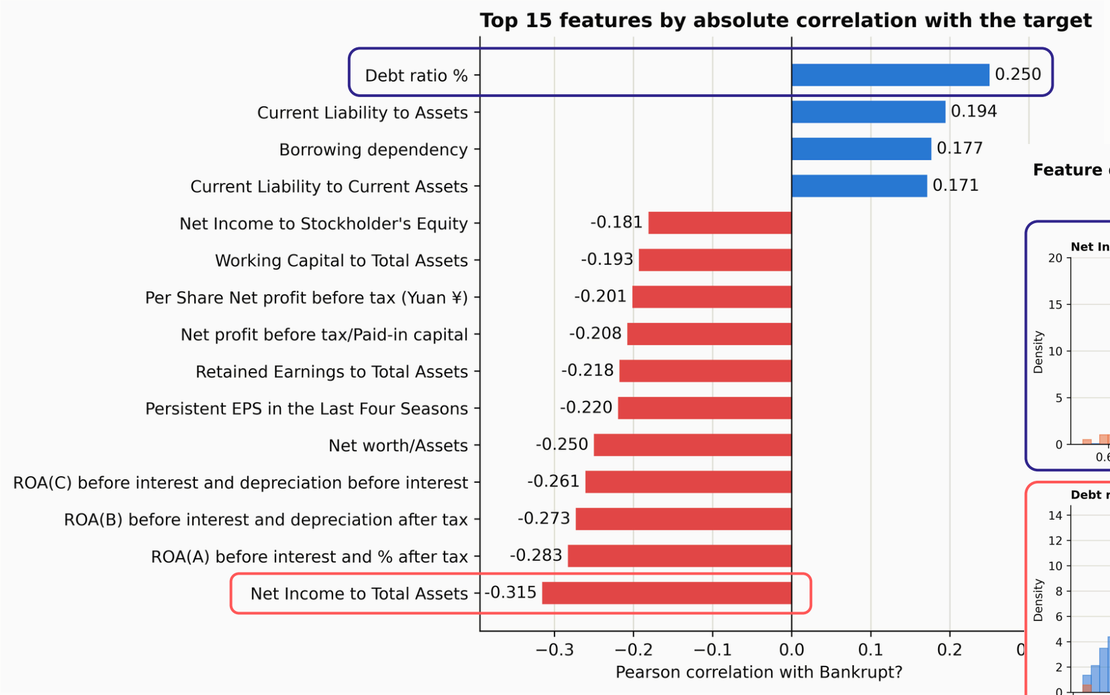
EXPLORATORY DATA ANALYSIS

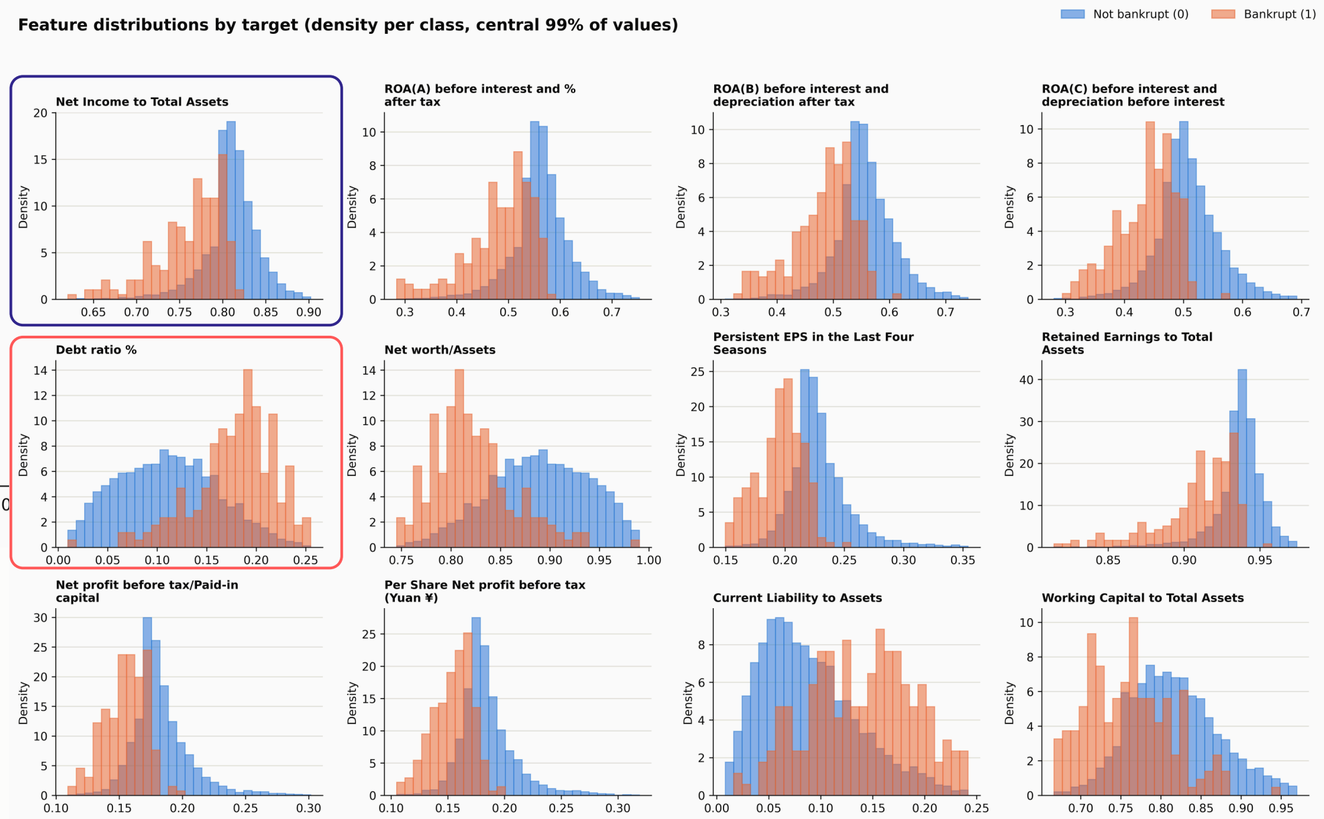
What does the data tell us before we build any model ?

We look at class balance and the financial ratios that separate bankrupt from healthy companies.



CORRELATION WITH THE TARGET

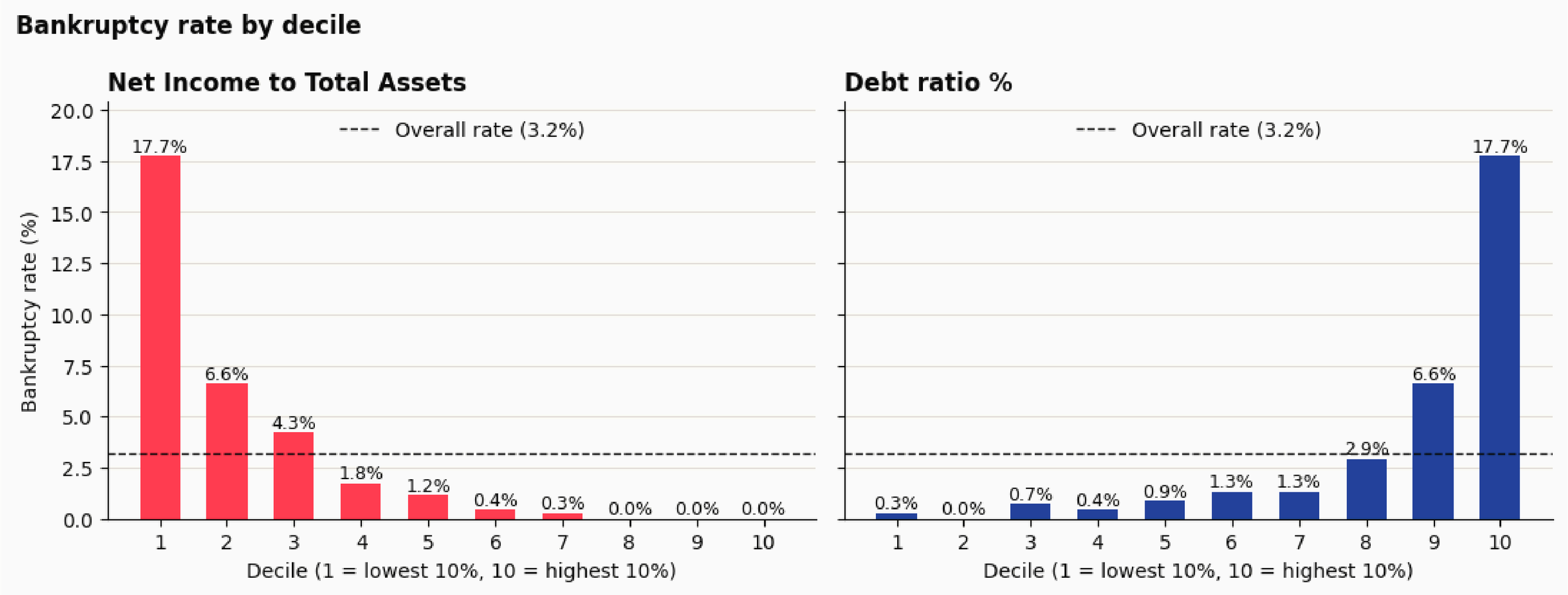




Low Profit + High Debt

“Bankruptcy Risk”

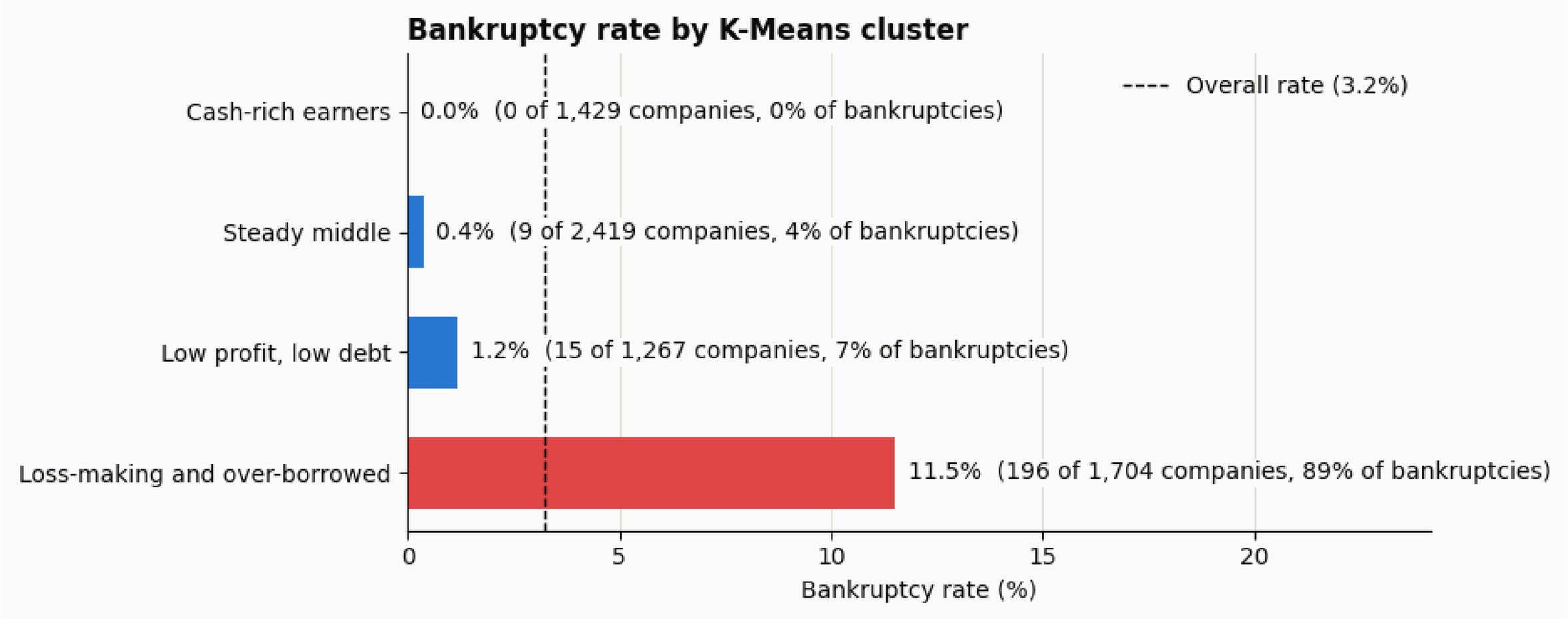
BUSINESS SEGMENTATION



BUSINESS SEGMENTATION

	Low Debt	High Debt
Low Profit	**Low profit, low debt** 1,267 companies	**Loss-making and over-borrowed** 1,704 companies
High Profit	**Cash-rish earner** 1,429 companies	**Steady middle** 2,419 companies

CLUSTER PROFILE



CLUSTER PROFILE



Cluster profiles (orange = above typical, blue = below typical)

	Cash-rich earners	Steady middle	Low profit, low debt	Loss-making and over-borrowed
Net Income to Total Assets	+1.07	+0.34	-0.71	-0.85
Retained Earnings to Total Assets	+1.02	+0.35	-0.69	-0.85
Debt ratio %	-0.85	+0.21	-0.91	+1.08
Borrowing dependency	-0.91	+0.12	-0.59	+1.10
Current Liability to Assets	-0.60	+0.22	-0.94	+0.89
Working Capital to Total Assets	+0.98	-0.14	+0.31	-0.86
Cash/Total Assets	+1.03	-0.15	+0.05	-0.69
Cash Flow to Total Assets	+0.71	-0.00	-0.40	-0.29

DATA PREPROCESSING

TRAIN / TEST SPLIT

Stratified split, 80 / 20

ALL COMPANIES 6,819 · 220 bankrupt



TRAINING SET 5,455 rows · 176 bankrupt

The model learns here: selection, outlier removal, SMOTE, tuning

TEST SET 1,364 rows · 44 bankrupt

Left untouched, used only for the final score

Not bankrupt Bankrupt **Same 3.2% bankrupt in both sets**

BASELINE

A default XGBoost, nothing added

- XGBoost with default hyperparameters
- Trained on all 95 features
- No outlier removal
- No resampling
- No tuning

ENGINEERED

Four steps, compared with the baseline

1. Feature selection (top 40 features)
2. IQR outlier removal (non-bankrupt rows only)
3. SMOTE (classes balanced 1:1 in training)
4. Tuned XGBoost

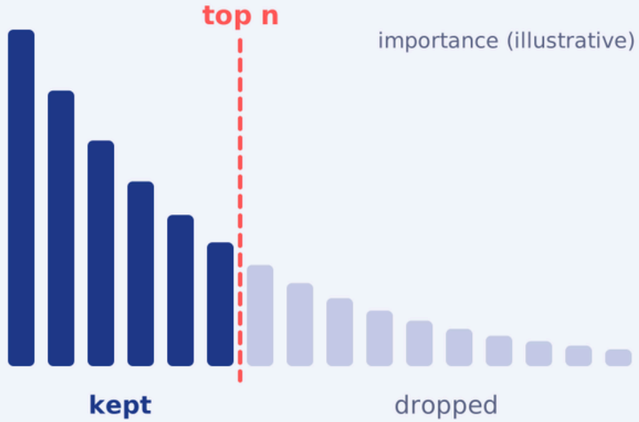


FEATURE ENGINEERING

- Preprocessing step for the engineered model

1 FEATURE SELECTION

Keep the top n features, ranked by importance in a preliminary XGBoost

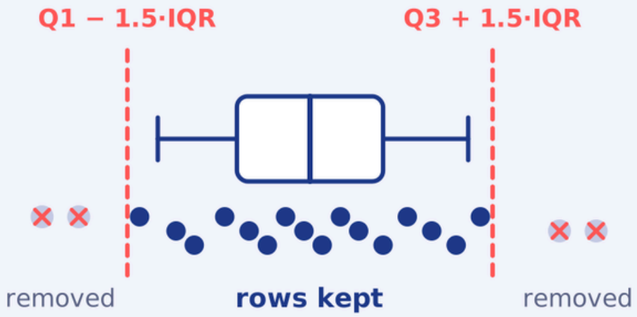


PRELIMINARY XGBOOST

200 trees · depth 3
scale_pos_weight = class ratio

2 IQR OUTLIER REMOVAL

Drop rows outside the fences
[Q1 − 1.5·IQR, Q3 + 1.5·IQR]

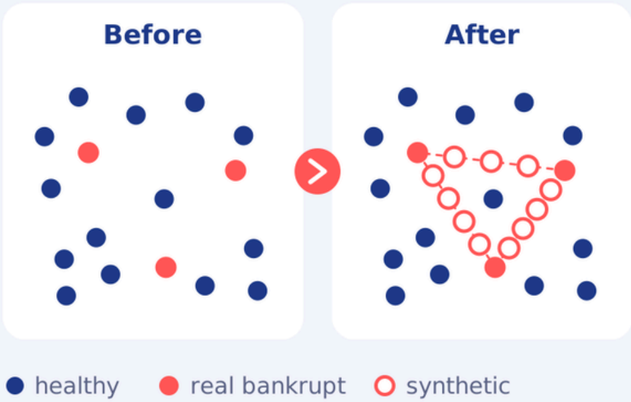


NON-BANKRUPT ROWS ONLY

Bankrupt rows are never removed:
219 of 220 contain an outlier

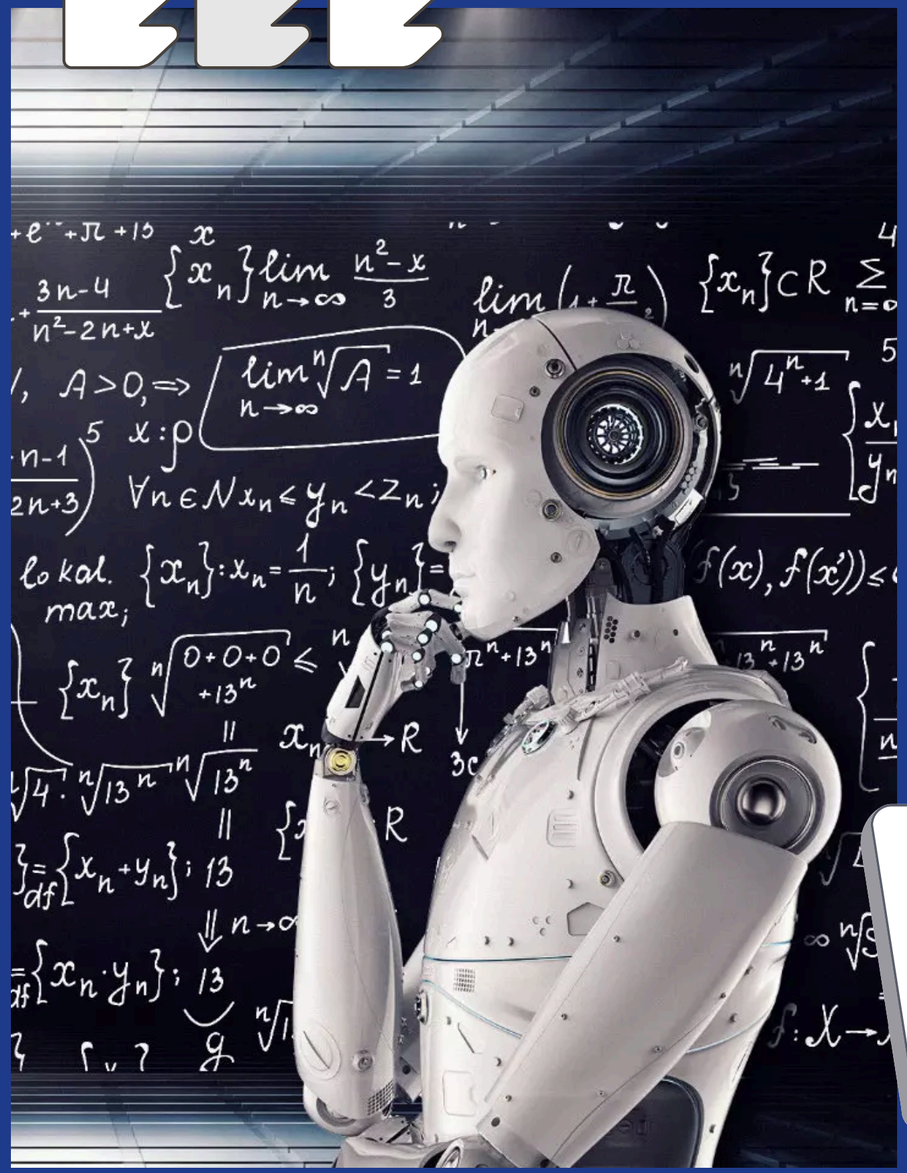
3 SMOTE

Interpolate between real bankrupt examples until the two classes are balanced 1:1



TRAINING DATA ONLY

176 real bankrupt rows stay;
synthetic ones fill the gap to 1:1



MODEL CREATION

BASELINE

All 95 features
no selection

→

XGBoost
default hyperparameters

✕ no outlier removal ✕ no resampling ✕ no tuning

ENGINEERED

Feature selection
top 40 features

→

IQR outlier removal
non-bankrupt rows only

→

SMOTE
classes balanced 1:1

→

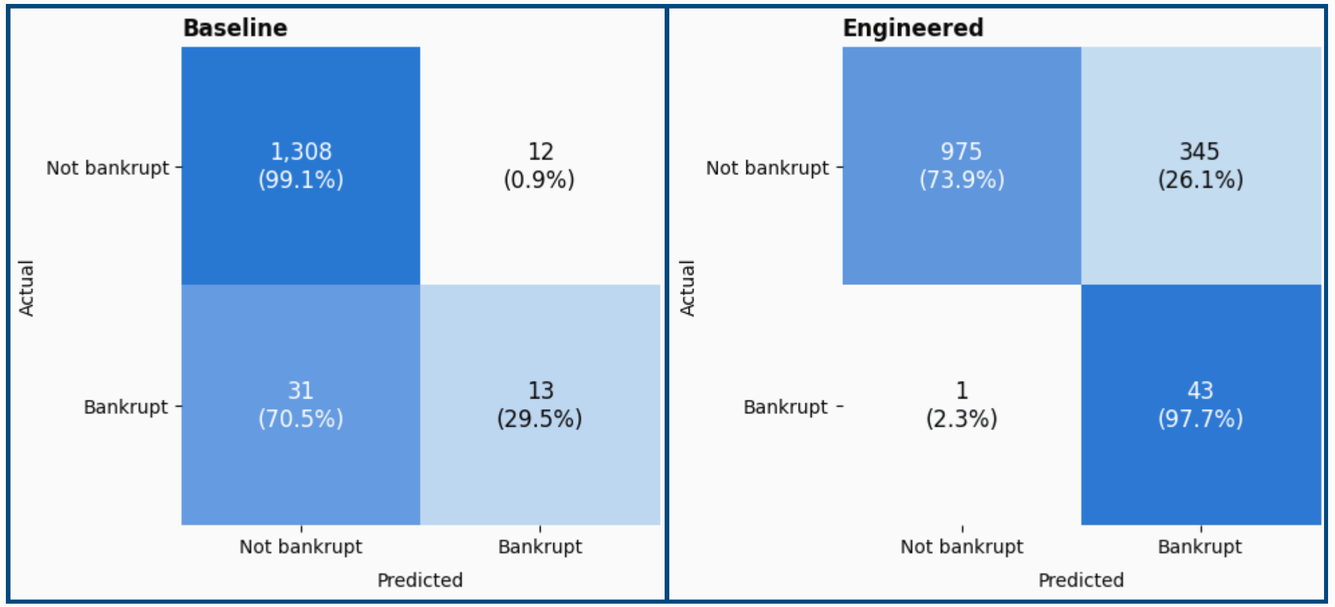
Tuned XGBoost
hyperparameter search



“catch as many bankrupt companies as possible”

ENGINEERED MODEL AND HYPERPARAMETER TUNING

Parameter	Search range	Best value
select__max_features (number of features)	10 – 40	40
clf__n_estimators	100 – 500	479
clf__learning_rate	0.01 – 0.2 (log scale)	0.0113
clf__max_depth	2 – 6	3
clf__min_child_weight	1 – 10	10
clf__subsample	0.6 – 1.0	0.643
clf__colsample_bytree	0.5 – 1.0	0.628



Model	Accuracy	Recall
Baseline	0.968	0.295
Engineered	**0.746**	**0.977**
Always "not bankrupt"	0.968	0

SUMMARY

WHAT WE FOUND

3.2% **of companies went bankrupt.**
Accuracy alone misleads: always saying “healthy” already scores 96.8%.

89% **of bankruptcies sit in one segment:**
low profit + high debt, which holds only 25% of all companies.

43/44 **bankrupt test companies caught**
by the engineered model, vs 13 for the baseline (recall 0.977 vs 0.295).
Trade-off: 345 healthy companies flagged too.

APPLY TO DASHBOARD



- **Screen a watchlist:** which segment does each holding resemble?
- **Check a company:** see how its closest peers fared.
- **Re-check before adding to a DCA position,** then read the financial statements.

Disclamer: a screening tool for investors and lenders, not a verdict. Taiwan data, 1999-2009. Educational, not investment advice.